Hydraulic standard cylinder type DZ25

operating pressure p_{max}	250 bar
lifting speed	max. 0.5 m/s



Product characteristics

- compact design
- with or without cushioning
- four different mounting styles

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Technical data

General

type	hydraulic cylinder, differential design, double-acting
mounting type	DZ25 M_ thread at cylinder head DZ25 G_ rod eye with spherical bearing at cylinder bottom DZ25 R_ round flange at cylinder head DZ25 S_ trunnion at cylinder head
corrosion protection	grounding with active pigmentation based on alkyd resins
ambient temperature	-30 to +70 °C
sealing type	S: standard seals R: low friction seals
tolerances	mounting dimensions: DIN2768-m-S/R stroke: DIN24333 piston rod: chromium plated layer thickness: 25±5 µm

Hydraulic parameters

Hydraulic fluid: mineral oil according to DIN 51524, other media on request

nominal pressure	250 bar
operating pressure	250 bar
statical test pressure	max. 20 bar
hydraulic fluid temperature	-25 to +70 °C
viscosity	10-600 mm²/s
permissible degree of pollution	max. class 22/19/16 according ISO 4406
filter recommendation	filter retention rate $\beta_{25} > 75$
lifting speed*	max. 0.5 m/s, higher speeds on request

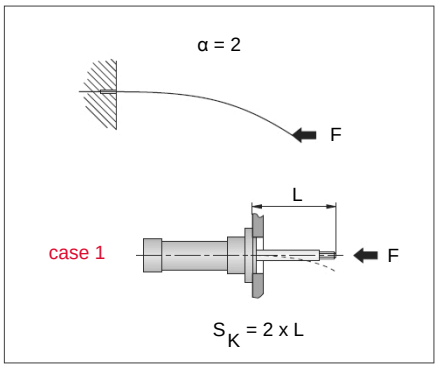
*At lifting speeds > 0.1 m/s the use of cylinders with end position cushioning is recommended. The cushioning process is not designed to brake huge external masses. This could considerably increase the system pressure. In this case confer with the factory.

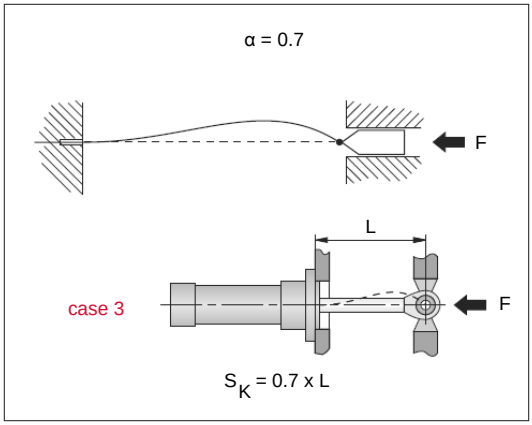
Mechanical parameters

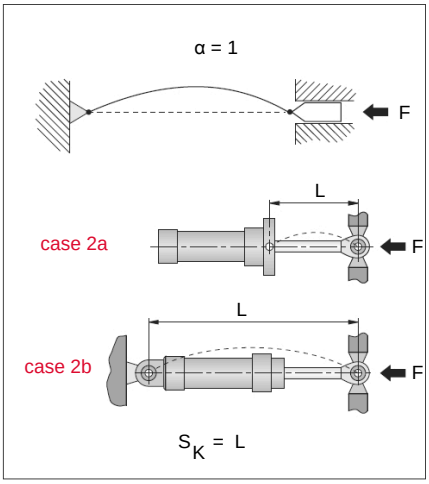
piston-∅ [mm]	32		40		50		63		80		100		125	
rod-∅ [mm]	18	22	22	28	28	36	36	45	45	56	56	70	70	90
piston area A [cm²]	8.1	8.1	12.6	12.6	19.7	19.7	31.2	31.2	50.3	50.3	78.6	78.6	122.8	122.8
annulus area A_R [cm²]	5.5	4.3	8.8	6.5	13.5	9.5	21.0	15.3	34.4	25.7	54.0	40.1	84.3	59.2
area ratio A/A_R	1.5	1.9	1.5	1.9	1.5	2.1	1.5	2.0	1.5	2.0	1.5	2.0	1.5	2.1
minimum stroke [mm]	45		45		45		45		50		60		65	
cushioning length [mm]	not possible		25		25		25		30		35		40	
damping ratio, head [cm²]	not possible		5.5		8.3		13.1		22.0		34.4		51.8	
damping ratio, bottom [cm²]	not possible		11.4		17.6		28.6		46.5		70.5		112.5	

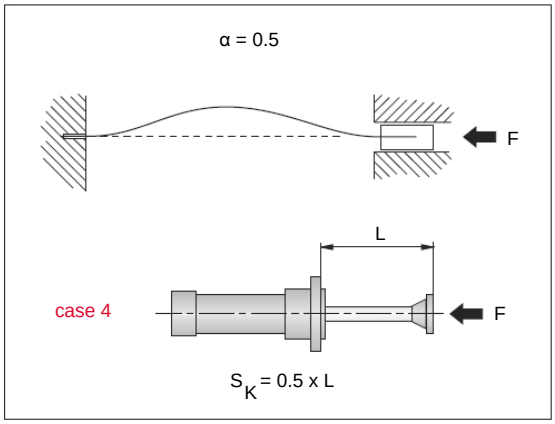
Determination of piston rod diameter

Load characteristics







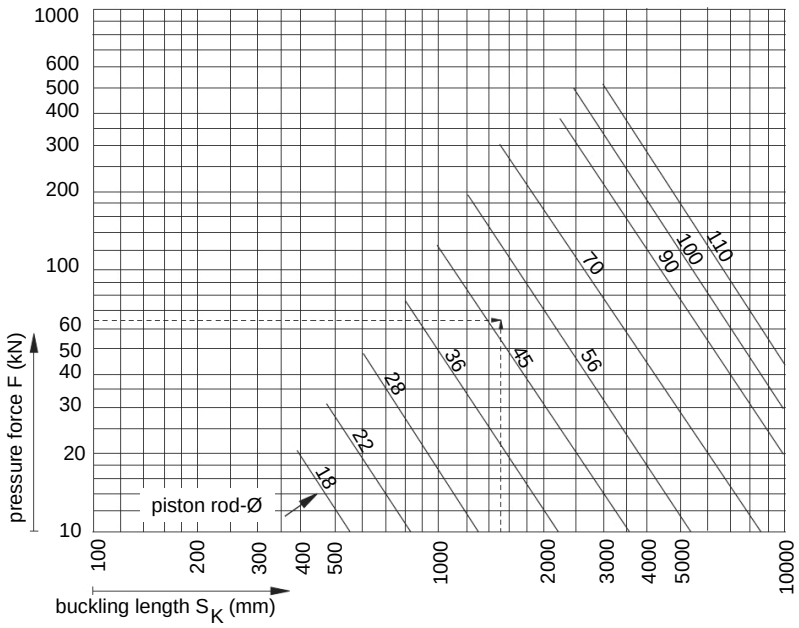


Case 4 is unfavourable and should be avoided.

1. Determination of load characteristic: α
2. Determination of the extended length: L
3. Calculation of the buckling length : $S_K = α \times L$
4. Determination of the rod diameter from the piston rod load diagram

Piston rod load diagram

buckling security = 3.5



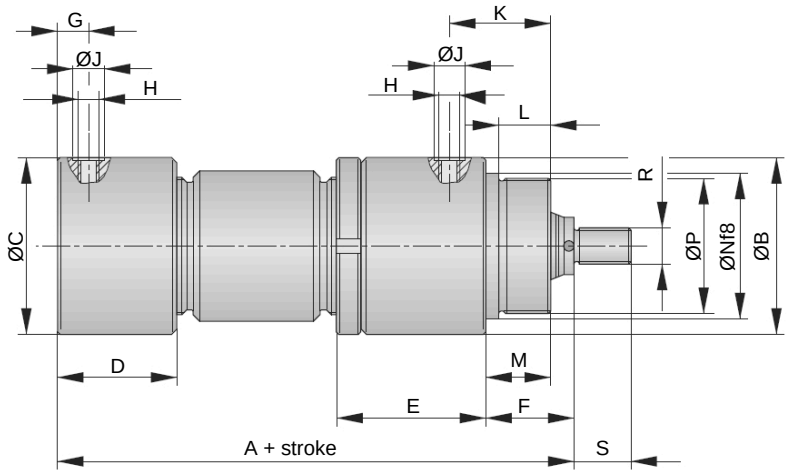
A cylinder according case 1 (α=2) with an extended length of 760 mm (Sk=1520 mm) results in a rod-Ø of 56 mm for a pressure force of 64 kN.

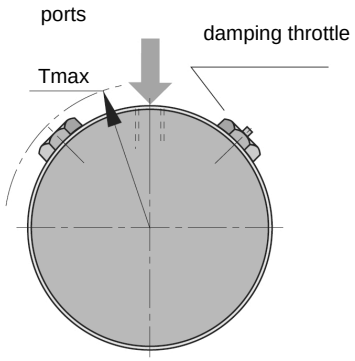
Example: For a cylinder according load characteristic 1 (→ α=2) with an extended length of 760 mm (→ S_K=1520 mm) a piston rod diameter of 56 mm arises from a pressure force of 64 kN.

Dimensions and connections

Dimensions are given in mm.

M: thread at cylinder head

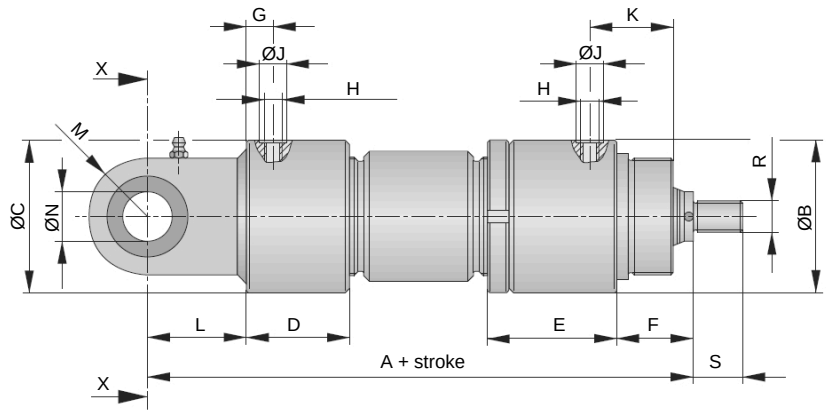


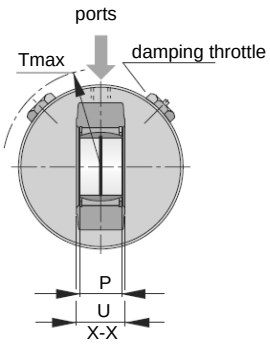


Dimensions dependent on piston diameter [mm]

piston ∅	32	40	50	63	80	100	125
rod ∅	18/22	22/28	28/36	36/45	45/56	56/70	70/90
A	121	143	155	171	187	208	241
B	56	67	77	92	112	138	172
C	56	67	77	90	108	133	167
D	57	63	66	75	80	89	100
E	72	83	85	88	99	108	123
F	33	37	42	50	57	67	80
G	15	16	16	21	21	23	27
H	G1/4	G3/8	G3/8	G1/2	G1/2	G1/2	G3/4
J	23	28	28	33	33	33	42
K	40	46	51	57	62	72	87
L	17	20	25	30	35	42	52
M	22	25	30	35	40	50	60
N	42	52	62	75	95	118	146
P	M40x1.5	M50x1.5	M60x1.5	M74x1.5	M94x2	M116x2	M145x2
R	M14x1.5	M16x1.5	M22x1.5	M28x1.5	M35x1.5	M45x1.5	M58x1.5
S	18	16	22	28	35	45	58
T	-	39	43	59	67	77	93

G: spherical eye at cylinder bottom

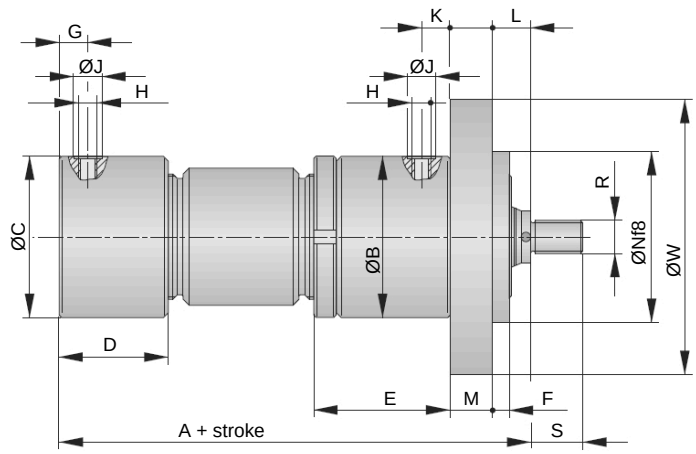


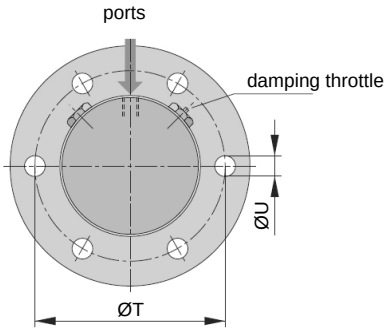


Dimensions dependent on piston diameter [mm]

piston ∅	32	40	50	63	80	100	125
rod ∅	18/22	22/28	28/36	36/45	45/56	56/70	70/90
A	157	188	206	232	256	296	341
B	56	67	77	92	112	138	172
C	56	67	77	90	108	133	167
D	57	63	66	75	80	89	100
E	72	83	85	88	99	108	123
F	33	37	42	50	57	67	80
G	15	16	16	21	21	23	27
H	G1/4	G3/8	G3/8	G1/2	G1/2	G1/2	G3/4
J	23	28	28	33	33	33	42
K	40	46	51	57	62	72	87
L	36	45	51	61	69	88	100
M	R20	R27.5	R32.5	R41.5	R50	R61.5	R70
N	16	25	30	35	40	50	60
P	16	20	22	25	28	35	44
R	M14x1.5	M16x1.5	M22x1.5	M28x1.5	M35x1.5	M45x1.5	M58x1.5
S	18	16	22	28	35	45	58
T	-	39	43	59	67	77	93
U	14	23	28	30	35	40	50

R: round flange at cylinder head

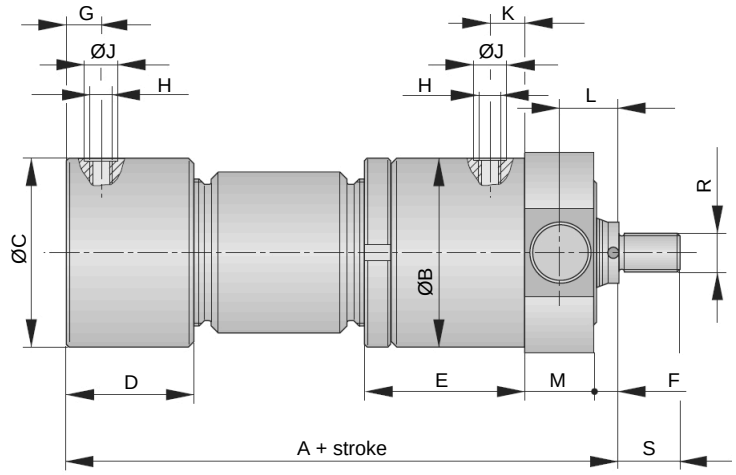


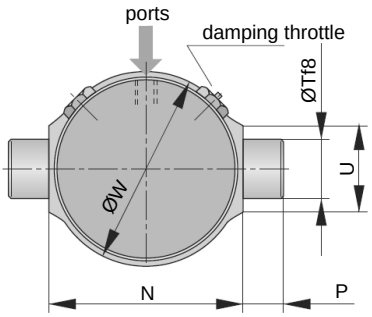


Dimensions dependent on piston diameter [mm]

piston ∅	32	40	50	63	80	100	125
rod ∅	18/22	22/28	28/36	36/45	45/56	56/70	70/90
A	121	143	155	171	187	208	241
B	56	67	77	92	112	138	172
C	56	67	77	90	108	133	167
D	57	63	66	75	80	89	100
E	72	83	85	88	99	108	126
F	4	4	4	4	4	6	6
G	15	16	16	21	21	23	27
H	G1/4	G3/8	G3/8	G1/2	G1/2	G1/2	G3/4
J	23	28	28	33	33	33	42
K	18	21	21	22	22	22	27
L	17	17	17	22	25	27	30
M	16	20	25	28	32	40	50
N	65	75	85	100	120	145	180
R	M14x1.5	M16x1.5	M22x1.5	M28x1.5	M35x1.5	M45x1.5	M58x1.5
S	18	16	22	28	35	45	58
T	82	95	105	125	150	175	215
U	9	11.5	11.5	14	18	18	22
W	100	120	130	155	185	210	260

S: trunnion at cylinder head



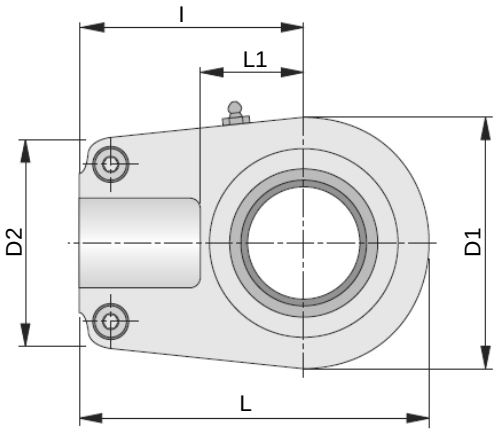


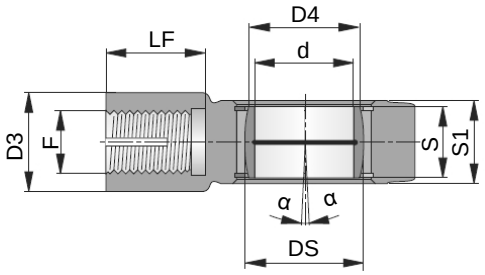
Dimensions dependent on piston diameter [mm]

piston ∅	32	40	50	63	80	100	125
rod ∅	18/22	22/28	28/36	36/45	45/56	56/70	70/90
A	121	143	155	171	187	208	241
B	56	67	77	92	112	138	172
C	56	67	77	90	108	133	167
D	57	63	66	75	80	89	100
E	71	80	85	87	96	105	120
F	12	13	13	16	18	18	21
G	15	16	16	21	21	23	27
H	G1/4	G3/8	G3/8	G1/2	G1/2	G1/2	G3/4
J	23	28	28	33	33	33	42
K	17	18	18	19	19	19	24
L	23	26,5	29	34,5	39	44	52
M	22	27	32	37	42	52	62
N	68	75	85	105	135	165	205
P	14	16	18	22	28	36	45
R	M14x1.5	M16x1.5	M22x1.5	M28x1.5	M35x1.5	M45x1.5	M58x1.5
S	18	16	22	28	35	45	58
T	20	25	30	35	40	50	60
U	25	35	40	45	50	60	75
W	64	72	85	105	130	155	195

Accessories

Spherical eyes

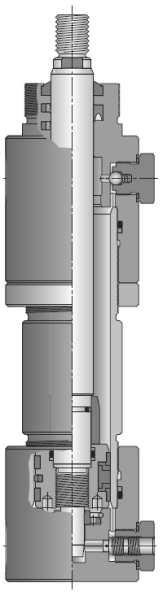




piston-∅	32	40	50	63	80	100	125
d	16^{H7}	$25_{-0.01}$	$30_{-0.01}$	$35_{-0.012}$	$40_{-0.012}$	$50_{-0.012}$	$60_{-0.015}$
l	44	50	60	70	85	105	130
S	16	20	22	25	28	35	44
LF	19	17	23	29	36	46	59
D1	40	56	64	78	94	116	130
D2	40	46	50	66	76	90	120
D3	21	25	32	40	49	61	75
D4	20	29	34	39,5	45	56	66,5
DS	23	35,5	40,7	47	53	66	80
S1	13	23	28	30	35	40	50
L	64	80	94	112	135	168	200
L1	18	28	30	38	45	55	65
F	M14x1.5	M16x1.5	M22x1.5	M28x1.5	M35x1.5	M45x1.5	M58x1.5
α	4°	7°	6°	6°	7°	6°	6°
ident. no.	HZ660142X	HZ660129X	HZ660130X	HZ660131X	HZ660132X	HZ660134X	HZ660135X

Seal kits

piston ∅	rod ∅	standard seals	low friction seals
32	18	HC01000	HC02000
32	22	HC01001	HC02001
40	22	HC01010	HC02010
40	28	HC01011	HC02011
50	28	HC01020	HC02020
50	36	HC01021	HC02021
63	36	HC01030	HC02030
63	45	HC01031	HC02031
80	45	HC01040	HC02040
80	56	HC01041	HC02041
100	56	HC01050	HC02050
100	70	HC01051	HC02051
125	70	HC01060	HC02060
125	90	HC01061	HC02061



Order information

Type code

DZ25	M	100/070	-	0800	B	M	S

- **sealing type**
- **rod ends**
- **cushioning****
- **stroke [mm]***
- **piston diameter / rod diameter**
- **mounting types**
- **type** hydraulic cylinder

mounting types

M	thread at cylinder head
G	rod eye with spherical bearing at cylinder bottom
R	round flange at cylinder head
S	trunnions at cylinder head

piston diameter / rod diameter

032	../018	../022
040	../022	../028
050	../028	../036
063	../036	../045
080	../045	../056
100	../056	../070
125	../070	../090

stroke [mm]*

...	freely selectable; In particular cases a support extension is necessary for extremely small cylinders (stroke > 15x piston diameter). Contact HAWE.

*check buckling stress when selecting

cushioning**

A	without cushioning
B	cushioning at the bottom
K	cushioning at the head
D	cushioning at both sides

**check buckling stress when selecting

rod ends

M	thread
G	rod eye with spherical bearing

sealing type

S	standard design
R	low-friction design (on request)

Contact details

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Phone (Customer Service)	+ 49 (0) 89 / 37 91 00 - 1491
Fax (Customer Service)	+ 49 (0) 89 / 37 91 00 - 91491
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